



MORTGAGE & LENDING

Payslip extraction two models must *agree* on

A production microservice for a UK mortgage broker that reads a payslip PDF, extracts every line into a strictly typed schema, and has two frontier models cross-check each other before anything reaches a human. Processing errors down 52%, manual review down 80%, and mortgage approvals 15% faster — across thousands of documents a week.

52%

FEWER PROCESSING ERRORS

80%

LESS MANUAL REVIEW

15%

FASTER MORTGAGE APPROVALS

1000_s

DOCUMENTS WEEKLY

THE SITUATION

Every mortgage waits on someone reading a payslip

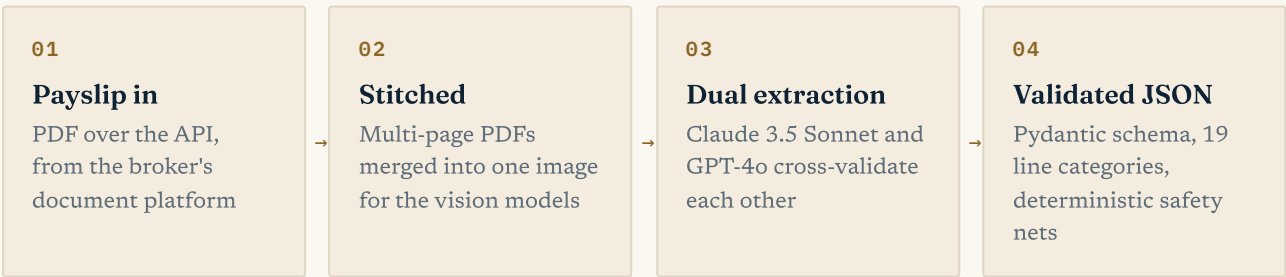
A UK mortgage broker processed thousands of payslips a week, and every one crossed a human desk. Staff read each document line by line – gross pay, tax, National Insurance, pension, the lot – and keyed the figures into the approval system by hand. Affordability decisions rested on that data, so it had to be right; but manual keying is slow and error-prone, and the review queue sat squarely in the critical path of every mortgage approval.

The brief was to take the keying out of the loop without taking the rigour out. A lending decision cannot rest on an unvalidated model guess, so “mostly right JSON” was never going to be acceptable. The service had to return data the broker's systems could trust – or say plainly that it couldn't.

WHAT I BUILT

Two models, one schema, no unchecked answers

The result is a FastAPI microservice – Dockerised, deployed on AWS – that accepts a payslip over a single endpoint and returns validated, structured JSON. Multi-page PDFs are stitched into one tall image so the vision models see the whole document at once, with nothing lost between pages. Two frontier models – Claude 3.5 Sonnet and GPT-4o – extract independently and cross-validate each other, and everything lands in a Pydantic schema with 19 typed payslip line categories, so a mislabelled deduction fails loudly instead of slipping through.



Around the models sits ordinary engineering discipline: deterministic post-processing that corrects the mistakes LLMs predictably make, retries on every model call, LangSmith tracing on every request, and Slack alerts when something needs a human. Quality was measured with an evaluation harness built for the job – model output scored by the exact number of edits a human reviewer would need to reach ground truth, which maps directly onto the reviewer labour the client was paying for.

19 TYPED PAYSリップ LINE CATEGORIES	15% FASTER MORTGAGE APPROVALS	52% FEWER PROCESSING ERRORS	80% LESS MANUAL REVIEW OVERHEAD
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THE HARD PARTS

What it takes to trust an LLM with lending data

A mortgage cannot rest on a guess

One model reading a payslip is an opinion; two agreeing is evidence. Claude 3.5 Sonnet and GPT-4o extract independently and cross-validate, with Pydantic validators enforcing the schema at the API boundary. Disagreements trigger a Slack alert rather than being quietly resolved.

Models make predictable mistakes

Vision models kept filing Tax and NI as deductions to sum, misreading negative adjustments and mangling name casing. A deterministic post-processing layer corrects each quirk in plain code – safety nets that hold regardless of which model sits behind them.

“Better” measured in human edits

Accuracy percentages hide what matters, so the harness counts the edits a reviewer needs to fix each extraction – one per wrong value, more for missing lines – with an LLM-as-judge pass so “Salary” and “Basic pay” aren't marked as errors. Changes were scored against baseline before shipping.

Architecture chosen by benchmark, not fashion

I built two full implementations – a LangChain single-chain extractor and a CrewAI multi-agent crew – and ran both through the same edit-count harness on real payslips. The evidence picked the winner; the client got the comparison, not just the conclusion.

THE RESULTS

PROCESSING ERRORS Down **52%**

MANUAL REVIEW Down **80%**

APPROVAL SPEED Mortgage approvals **15%** faster

IN PRODUCTION Thousands of documents weekly; client anonymised under NDA

WHAT THE CLIENT SAID

“Adam worked on an AI project with us and quickly demonstrated good technical skills, producing logical and well-structured code... he showed a strong commitment to delivering measurable results and consistently approached his work with a high level of professionalism. ... He would be a valuable asset to any development team.”

UPWORK CLIENT REVIEW · SENIOR MACHINE LEARNING ENGINEER ENGAGEMENT · 4.9★ · \$12,212.50 · 163 HOURS

ABOUT ADAM

Freelance AI engineer – **Expert-Vetted on Upwork (top 1%)**, 100% Job Success over 70+ projects, \$400K+ earned, 5,750+ hours billed. I build production LLM systems for regulated industries: insurance, finance, law.

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